

Assignment A1

MATH 147

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1 Assignment A1

1.1 Problem 1

(I) **Prove that $\sqrt{3}$ is irrational.**

We need to prove a lemma first before we prove the proposition.

Lemma 1. *For all $n \in \mathbb{Z}$, if $3 \mid n^2$, then $3 \mid n$.*

Proof. Suppose $3 \nmid n$. There are two cases to discuss.

case 1: $n = 3k + 1$

According to the definition of divisibility, we have

$$n = 3k + 1 \quad \exists k \in \mathbb{Z}.$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3c + 1 \quad c = 3k^2 + 2k \in \mathbb{Z}.$$

thus, $3 \nmid n^2$.

case 2: $n = 3k + 2$

According to the definition of divisibility, we have

$$\exists k \in \mathbb{Z} \text{ s.t. } n = 3k + 2$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3c + 1 \quad c = 3k^2 + 4k + 1 \in \mathbb{Z}.$$

thus, $3 \nmid n^2$. ■

Here is the proof of the question:

Proof. Suppose $\sqrt{3} \in \mathbb{Q}$.

Then, according to the definition of rational numbers,

$$\exists m, n \in \mathbb{Z}, n \neq 0 \quad \text{s.t.} \quad \sqrt{3} = \frac{m}{n}.$$

We assume that m and n have no common factor other than 1. Taking the square of the both sides, we get

$$3 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 3n^2,$$

which implies that $3 \mid m$ according to the lemma 1. According to the definition of divisibility,

$$\exists k \in \mathbb{Z} \quad \text{s.t.} \quad m = 3k.$$

Substituting $m = 3k$ back into $m^2 = 3n^2$, we have

$$\begin{aligned} n^2 &= (3k)^2 / 3 \\ &= 9k^2 / 3 \\ &= 3d, d = k^2 \in \mathbb{Z}. \end{aligned}$$

It means that $3 \mid n$.

Since $3 \mid n$ and $3 \mid m$, it contradicts the assumption that n and m have no common factor other than 1.

Therefore, $\sqrt{3} \in \mathbb{R} \setminus \mathbb{Q}$. ■

(II) *Proof.* We define

$$P := \{x \in \mathbb{R} : x > 0\}$$

to be a nonempty subset of $\mathbb{R}$ that satisfies Trichotomy law, Closure under addition, and closure under multiplication.

Suppose $a, b \in P$. There are two cases to discuss, which are

- $a > b$.
- $a = b$.
- $b < a$.

We will prove case $a > b$ first.

According to the given statement, we have

$$(\forall c > 0) (\exists x > 0) x^2 = c.$$

It follows that there $\exists a', b' \in P$ such that

$$(a')^2 = a \quad \text{and} \quad (b')^2 = b.$$

Since $a > b$, it means that

$$\begin{aligned} a - b &= (a')^2 - (b')^2 \\ &= (a' - b')(a' + b') \in P. \end{aligned}$$

We know that $a' + b' \in P$, so $a' - b' \in P$.

By the ordered field properties, we have

$$(a' - b')^2 \in P \implies (a' - b')^2 > 0.$$

given $a' - b' \in P$ and closure under multiplication.

Expanding the left side, we get

$$(a')^2 - 2a'b' + (b')^2 > 0 \implies \frac{(a')^2 + (b')^2}{2} > a'b'. \quad (1.1)$$

By substituting $(a')^2 = a$ and $(b')^2 = b$ into equation (1.1), we have

$$\frac{a + b}{2} > a'b'.$$

Since $a' \in P$ and $b' \in P$, it follows from the closure under multiplication that

$$a'b' \in P \implies a'b' > 0.$$

Moreover,

$$(a'b')^2 = (a')^2 (b')^2 = ab \in P \implies ab > 0$$

Since $a'b' > 0$ and $ab = a'b'$, therefore, ab is a positive square root of $a'b'$.

$$a'b' = \sqrt{ab}.$$

is the only option.

As a result,

$$\frac{a+b}{2} > \sqrt{ab}.$$

For $b > a$, it can be proved using similar way as $a > b$ by swapping a with b and b with a .

For $a = b$, we have

$$\frac{a+b}{2} = \frac{2a}{2} = a = \sqrt{a^2} = \sqrt{ab}.$$

Therefore, we get

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Thus, the inequality has been proved. ■

Reference: The ordered field properties were consulted in V. A. Zorich, *Mathematical Analysis I*, Section 2.2. Prove style is modified based on the suggestions from my dear father, Changwei Hu.

1.2 Problem 2

1.2.1 (I)

Definition 1. A set $S \subseteq \mathbb{R}$ is called inductive if $1 \in S$ and $k \in S$ implies $k + 1 \in S$.

$\mathbb{R}$ is inductive since $1 \in \mathbb{R}$ and for all $x \in \mathbb{R}$, $x + 1 \in \mathbb{R}$.

$(0, \infty)$ is inductive since $1 \in (0, \infty)$ and for all $x \in (0, \infty)$, $x + 1 \in (0, \infty)$.

$(0, \infty) \setminus \{5\}$ is not inductive since $4 \in (0, \infty) \setminus \{5\}$ but $4 + 1 = 5 \notin (0, \infty) \setminus \{5\}$.

1.2.2 (II)

Proof. Suppose that S and T are inductive sets, and $U = S \cap T$.

By the definition of inductive set, we have

$$1 \in S \quad \text{and} \quad k \in S \implies k + 1 \in S. \quad (1.2)$$

and

$$1 \in T \quad \text{and} \quad k \in T \implies k + 1 \in T. \quad (1.3)$$

Since 1 is in both S and T , it means that $1 \in U$.

By statements (1.2) and (1.3), we get

$$k \in U \iff (k \in S \wedge k \in T) \implies (k + 1 \in S \wedge k + 1 \in T) \iff k + 1 \in U.$$

Thus, U is an inductive set. ■

1.2.3 (III)

Proof. Let $\mathcal{I}$ be the collection of all inductive subsets. We define

$$\mathbb{N} = \bigcap_{S \in \mathcal{I}} S.$$

Since

$$(\forall S \in \mathcal{I}) 1 \in S,$$

it follows that $1 \in \mathbb{N}$.

It suffices to show that

$$k \in \mathbb{N} \implies k + 1 \in \mathbb{N}.$$

Suppose $k \in \mathbb{N}$. By the definition of the intersection of sets, it means that

$$(\forall S \in \mathcal{I}) k \in S.$$

It follows the definition of inductive set that

$$(\forall S \in \mathcal{I}) k + 1 \in S,$$

which implies

$$k + 1 \in \mathbb{N}.$$

Thus, $\mathbb{N}$ is inductive. ■

1.3 Problem 3

1.3.1 (I)

Proof. Suppose that $P(1)$ is true and

$$(\forall n \in \mathbb{N}) [(P(1) \wedge P(2) \wedge \cdots \wedge P(n)) \implies P(n+1)].$$

Let

$$T(n) := P(1) \wedge P(2) \wedge \cdots \wedge P(n).$$

Since $P(1)$ is true, it implies that $T(1)$ is also true.

By the assumption of strong induction, we obtain

$$T(n) \implies P(n+1),$$

and hence, since $T(n)$ is already assumed,

$$T(n) \wedge P(n+1).$$

By the definition of $T(n+1)$, it follows that

$$T(n) \implies T(n+1).$$

Following the Principle of Mathematical Induction, this yields

$$(\forall n \in \mathbb{N}) T(n) \implies (\forall n \in \mathbb{N}) P(n).$$

Thus, Principle of Strong Mathematical Induction is correct. ■